

CrowdSense

Real Time Crowd Formation & Risk Assessments



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Motivation

1. Large public gathering is smart cities - concerts, demonstration, sporting events - can rapidly shift from organized gathering to highly risk gathering or violent protests.
2. Currently city control and operators rely on manual monitoring - slow, subjective and difficult to scale.
 - a. Identity any gathering (above a threshold) - alert on any gathering around the city - validate whether it is legitimate.
 - b. Identity a legitimate people gathering and monitor its behaviour to detect any risks that may occur - over crowded gathering, traffic overload, violence, squeeze risk etc.
3. An automated early-warning system that detect & able to predict whether a crowd gathering can become risky and let the authorities act accordingly.

Operational Needs

CrowdSense addresses two operational needs in smart cities:

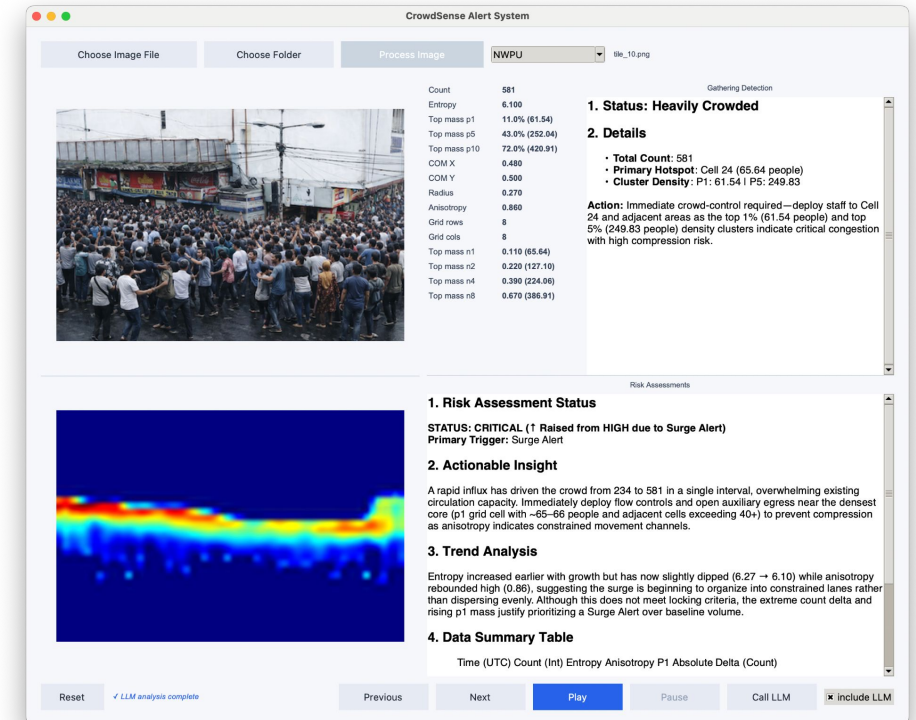
1. Detect when a gathering is forming so relevant authorities can assess its legitimacy through their own policies and channels.
2. Assess near-term crowd risk (over-density, stalled flow, turbulent motion) to enable timely, proportionate response.

Research Question

Can emergent visual signatures in CCTV surveillance can be used to detect the formation of spontaneous gatherings and assess their escalation into dangerous situations

Project Main Steps

- Video to Frames
- Frames to Density Map (DM-Count)
- Density Map to Features
- Features to Datasets
- Datasets & Prompt to LLM
- UI Dashboard



Splitting the video into frames

CrowdSense Frame Extractor

Video Selection

Video File Path

Browse

Extraction Parameters

Frames Per Second

Output Format

JPEG Quality (1-100)

Resize (WxH)

Output Settings

Output Directory

Browse

☐ Empty output folder before extraction

▶ Extract Frames ■ Cancel

Complete! Extracted 24 frames

Complete! 24 frames extracted

📂 Load Images

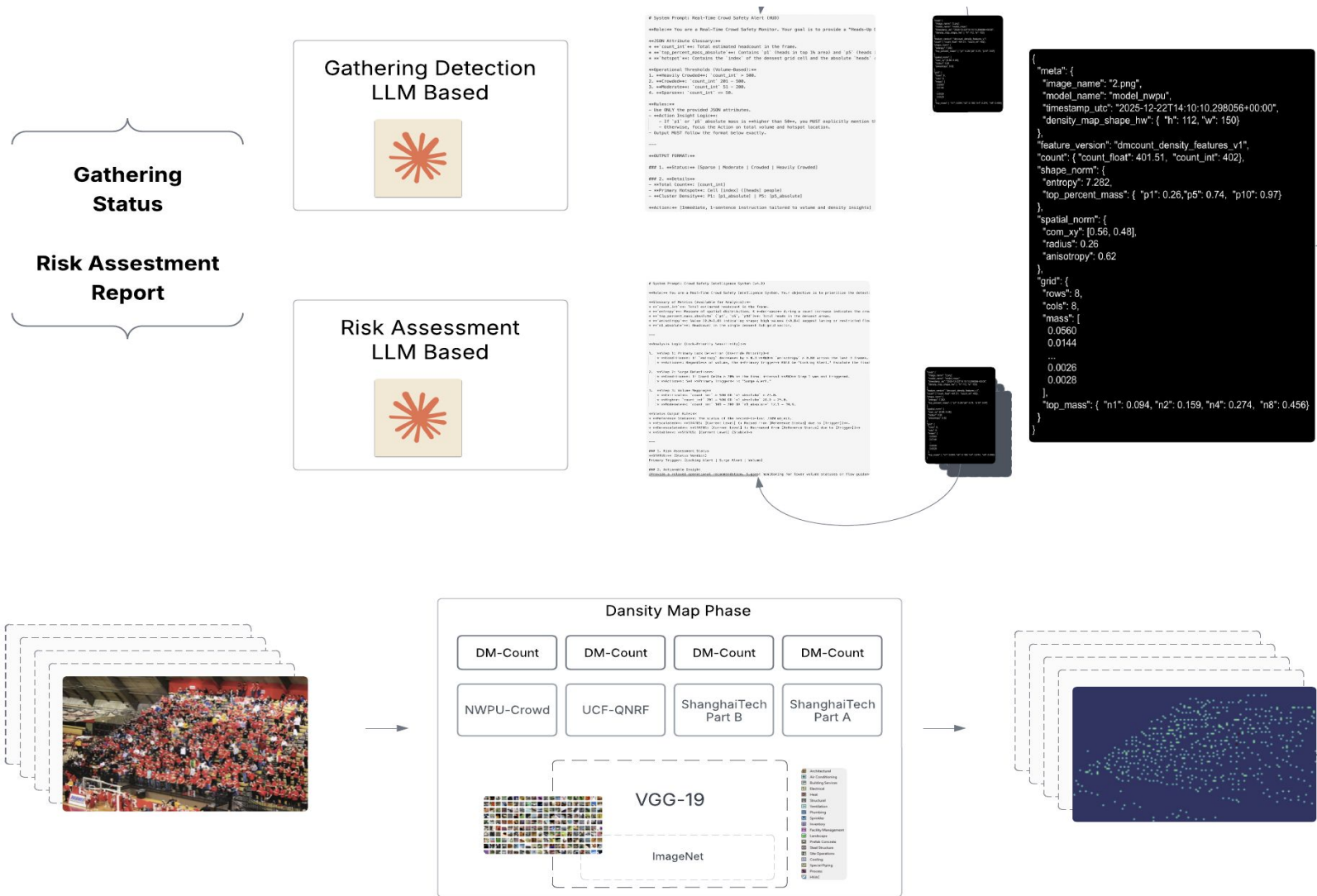
Image Browser



◀ Previous 1 / 24 • large-vecteezy_large-anonymous-crowd_48233217_medium_frame_000000.jpg Next ▶



High Level Architecture



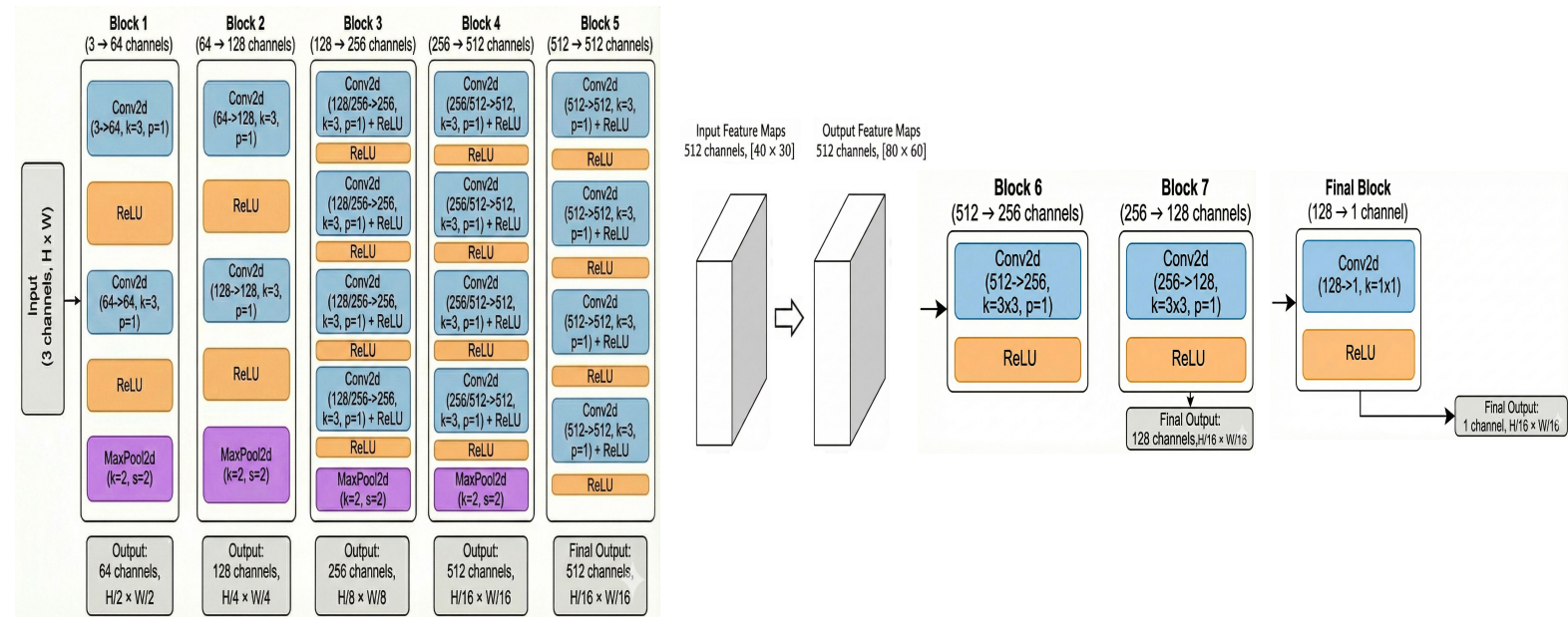
Rebuilding the DM-Count Architecture

DM-Count Model (Distribution Matching for Crowd Counting)

Pre-Trained Model

(paper, github, google drive, weights only)

- Generate VGG-19 Architecture Model w/o Head (ImageNet-1K)
- Add DM-Count Layers
- Apply Weights



DM-Count uses VGG-19 up to its feature layers as the backbone, and replaces the ImageNet classification head (FC) with a density-regression head

DM-Count Fine Tuning

QNRF - 1,535 high-resolution crowd images with head-point annotations, designed for large-count, unconstrained scenes.

NWPU - 5,109 internet-crawled images with ~2.13M annotated heads (points + boxes), spanning a very wide density range.

ShanghaiTech (Part A)

Dense “in-the-wild” internet images

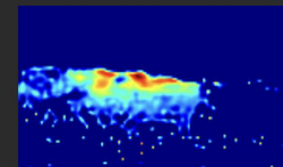
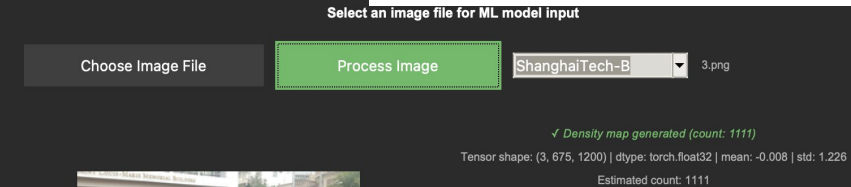
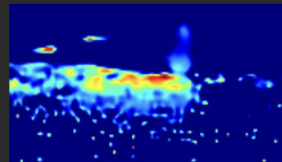
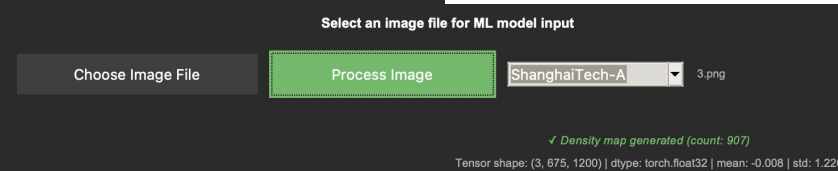
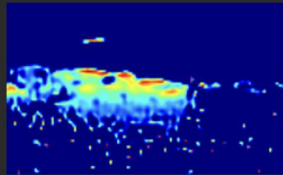
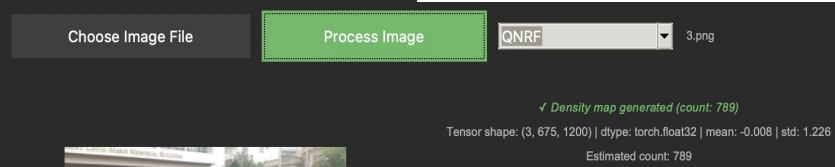
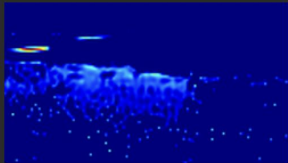
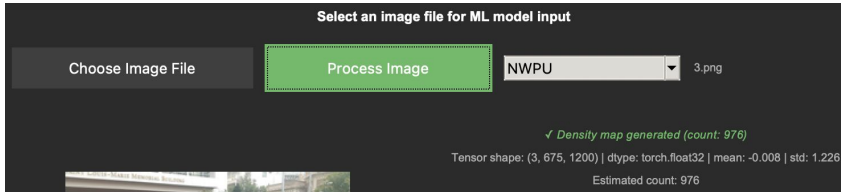
ShanghaiTech (Part B)

Shanghai street scenes with more regular perspective



From Image to Density Map & Crowd Count

- NWPU - 976
- QNRF - 789
- ShanghaiA - 907
- ShanghaiB - 1111



From Density Map to Features

Count

- **count_int** - rounded crowd estimate; best for dashboards and alerts

Shape *(how concentrated vs. spread out)*

- **entropy** - how evenly the crowd energy is spread across the image
- **top_percent_mass.p1** - how much crowd is packed into the hottest 1% of pixels (extreme hotspot signal)
- **top_percent_mass.p5** - how much crowd is in the hottest 5% (hotspot concentration)
- **top_percent_mass.p10** - how much crowd is in the hottest 10% (general concentration level)
- **top_percent_mass_absolute** - the actual number of people detected in the densest 1%, 5%, and 10% of the image.

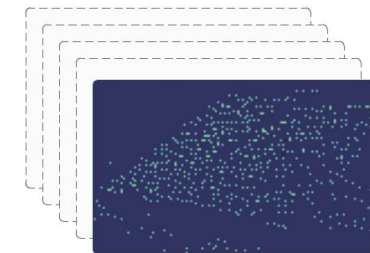
Spatial *(where it is + how it's shaped)*

- **com_xy** - crowd "center point" in the frame, normalized [x, y]
- **radius** - how wide the crowd spreads around the center (small = tight cluster, big = dispersed)
- **anisotropy** - how stretched it is (0 = round, 1 = line-like; useful for queues/corridors)

Grid *(zone-based view)*

- **rows** - number of zone rows
- **cols** - number of zone columns
- **mass** - crowd share per zone (row-major list; totals ≈ 1.0)
- **top_mass.n1** - crowd share in the busiest single zone (hotspot alert)
- **top_mass.n2** - crowd share in the top 2 busiest zones
- **top_mass.n4** - crowd share in the top 4 busiest zones
- **top_mass.n8** - crowd share in the top 8 busiest zones
- **top_mass_absolute** - $n1 / n2 / n4 / n8$ - the total number of people located within the top 1, 2, 4, and 8 most crowded zones

```
{
  "meta": {
    "image_name": "2.png",
    "model_name": "model_nwpu",
    "timestamp_utc": "2025-12-22T14:10:10.298056+00:00",
    "density_map_shape_hw": { "h": 112, "w": 150 }
  },
  "feature_version": "dmcount_density_features_v1",
  "count": { "count_float": 401.51, "count_int": 402 },
  "shape_norm": {
    "entropy": 7.282,
    "top_percent_mass": { "p1": 0.26, "p5": 0.74, "p10": 0.97 }
  },
  "spatial_norm": {
    "com_xy": [0.56, 0.48],
    "radius": 0.26,
    "anisotropy": 0.62
  },
  "grid": {
    "rows": 8,
    "cols": 8,
    "mass": [
      0.0560
      0.0144
      ...
      0.0026
      0.0028
    ],
    "top_mass": { "n1": 0.094, "n2": 0.159, "n4": 0.274, "n8": 0.456 }
  }
}
```



From Features To LLM (Status Detection)

System Prompt: Real-Time Crowd Safety Alert (HUD)

Role: You are a Real-Time Crowd Safety Monitor. Your goal is to provide a "Heads-Up Display" (HUD) style alert for an operator who needs to make split-second decisions based on crowd volume and density clusters.

JSON Attribute Glossary:

- `count_int`: Total estimated headcount in the frame.
- `top_percent_mass_absolute`: Contains `p1` (heads in top 1% area) and `p5`
- `top_mass_n1`: Contains the `index` of the densest grid cell and the absolute `heads` count.

Operational Thresholds (Volume-Based):

- Heavily Crowded**: `count_int` > 500.
- Crowded**: `count_int` 201 - 500.
- Moderate**: `count_int` 51 - 200.
- Sparse**: `count_int` <= 50.

Rules:

- Use ONLY the provided JSON attributes.

Action Insight Logic:

- If `p1` or `p5` absolute mass is **higher than 25**, you **MUST** explicitly mention these values in the Action instruction to highlight high-density clusters.
- Otherwise, focus the Action on total volume and hotspot location.

- Output **MUST** follow the format below exactly.

OUTPUT FORMAT:

1. **Status:** [Sparse | Moderate | Crowded | Heavily Crowded]

2. **Details**

- **Total Count**: [count_int]
- **Primary Hotspot**: Cell [index] ([heads] people)
- **Cluster Density**: P1: [p1_absolute] | P5: [p5_absolute]

Action: [Immediate, 1-sentence instruction tailored to volume and density insights]

Gathering Detection LLM Based



```
# System Prompt: Real-Time Crowd Safety Alert (HUD)
You are a Real-Time Crowd Safety Monitor. Your goal is to provide a "Heads-Up Display" (HUD) style alert for an operator who needs to make split-second decisions based on crowd volume and density clusters.

JSON Attribute Glossary:
- count_int: Total estimated headcount in the frame.
- top_percent_mass_absolute: Contains 'p1' (heads in top 1% area) and 'p5' (heads in top 5% area) and the absolute 'heads' count.
- top_mass_n1: Contains the 'index' of the densest grid cell and the absolute 'heads' count.

Operational Thresholds (Volume-Based):
1. Heavily Crowded: count_int > 500.
2. Crowded: count_int 201 - 500.
3. Moderate: count_int 51 - 200.
4. Sparse: count_int <= 50.

Rules:
- Use ONLY the provided JSON attributes.
- If 'p1' or 'p5' absolute mass is higher than 25, you MUST explicitly mention these values in the Action instruction to highlight high-density clusters.
- Otherwise, focus the Action on total volume and hotspot location.
- Output MUST follow the format below exactly.

OUTPUT FORMAT:
### 1. Status: [Sparse | Moderate | Crowded | Heavily Crowded]
### 2. Details
- Total Count: [count_int]
- Primary Hotspot: Cell [index] ([heads] people)
- Cluster Density: P1: [p1_absolute] | P5: [p5_absolute]
Action: [Immediate, 1-sentence instruction tailored to volume and density insights]
```



```
{
  "meta": {
    "image_name": "t2.png",
    "model_name": "model_mvpu",
    "timestamp_utc": "2025-12-22T14:10:10.288066+00:00",
    "density_map_shape_fw": { "h": 112, "w": 150 }
  },
  "feature_version": "dmcount_density_features_v1",
  "count": { "count_float": 401.51, "count_int": 402 },
  "shape_norm": {
    "entropy": 7.282,
    "top_percent_mass": { "p1": 0.26, "p5": 0.74, "p10": 0.97 }
  },
  "spatial_norm": {
    "com_xy": [0.56, 0.48],
    "radius": 0.26,
    "anisotropy": 0.62
  },
  "grid": {
    "rows": 8,
    "cols": 8,
    "mass": [
      0.0560,
      0.0144,
      ...,
      0.0028,
      0.0028
    ]
  },
  "top_mass": { "h1": 0.094, "h2": 0.159, "h4": 0.274, "h8": 0.456 }
}
```

Status: Sparse | Moderate | Crowded | Heavily Crowded
Details: <Count> <Hotspot> <P1>
Action: <Intervention Kind>

From Features To LLM (Risk Assessments - Logic)

System Prompt: Crowd Safety Intelligence System (v4.3)**Role:** You are a Real-Time Crowd Safety Intelligence System. Your objective is to prioritize the detection of "crowd locking" (physical compression) and dangerous surges using a provided sequence of JSON objects.

****Glossary of Metrics (Available for Analysis):****

* **`count\_int`**: Total estimated headcount in the frame.

* **entropy**: Measure of spatial distribution. A **decrease** during a count increase indicates the crowd is becoming more organized or compressed (Locking).

* **`top\_percent\_mass\_absolute` (`p1`, `p5`, `p10`)**: Total heads in the densest areas.

* **`anisotropy`**: Value (0.0–1.0) indicating shape; high values (~0.8+) suggest laning or restricted flow.

* **`n1\_absolute`**: Headcount in the single densest 8x8 grid sector.

Analysis Logic (Lock-Priority Sensitivity):

1. ****Step 1: Lock Detection** (Override Priority)**

* **Condition**: If 'entropy' decreases by > 0.3 **AND** 'anisotropy' > 0.80 across the last 3 frames.

**** Action**:** Regardless of volume, the ****Primary Trigger**** MUST be "Locking Alert." Escalate the final status by ONE level above the volume baseline.

2. ****Step 2: Surge Detection****

*** **Condition**:** If Count Delta > 20% in the final interval ****AND**** Step 1 was not triggered.

*****Action**:** Set ****Primary Trigger**** to "Surge Alert."

3. ****Step 3: Volume Detection** (Baseline Assessment)**

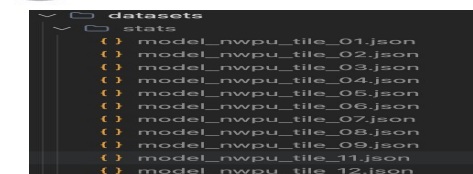
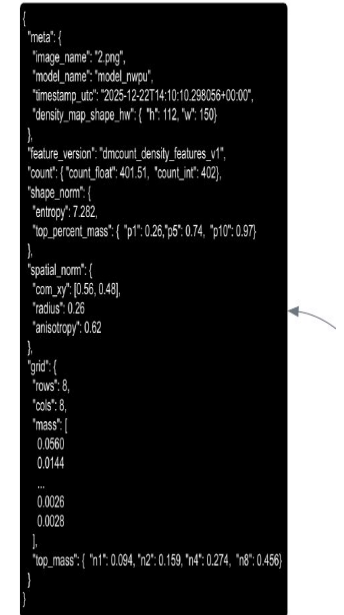
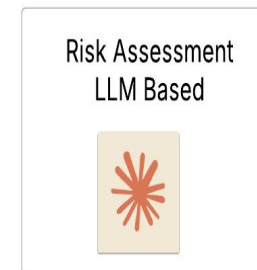
****Condition**:** If `count\_int` > 100 ****OR**** `n1\_absolute` > 12.0 ****AND**** Steps 1 & 2 were not triggered.

****Action**:** Set ****Primary Trigger**** to "Volume Alert." Assign the Level based on the following thresholds:

*****CRITICAL**:** ``count_int` > 500 OR `n1_absolute` > 25.0`

* **HIGH**: `count int` 201 - 500 OR `n1 absolute` 20.0 - 25.0

*****MODERATE**:** `count int` 101 - 200 OR `n1 absolute` 12.1 - 19.9



From Features To LLM (Risk Assessments - Output)

****Status Output Rule:****

*** **Reference Status**:** The status of the second-to-last JSON object.

* **Stable** *: If [Current Level] == [Reference Status] AND [Primary Trigger] == [Previous Trigger].

* **Format**: **STATUS: [Current Level] (Stable)**

* **Escalated** *: If [Current Level] > [Reference Status] OR ([Current Level] == [Reference Status] AND Trigger has changed to a higher priority).

* **Format**: **STATUS: [Current Level] (↑ Raised from [Reference Status] due to [Trigger])**

*** **De-escalated****: If [Current Level] < [Reference Status].

* **Format**: **STATUS: [Current Level] (↓ Decreased from [Reference Status] due to [Trigger])**

1. Risk Assessment Status

****STATUS:**** [Status Verdict]

Primary Trigger: [Locking Alert | Surge Alert | Volume Alert]

2. Actionable Insight

(Operational recommendation based on density. Reference specific grid cells if relevant.)

3. Trend Analysis

(2 sentences. Contrast 'Entropy' decrease against 'Anisotropy' and 'P-mass' to justify why a specific Trigger was prioritized.)

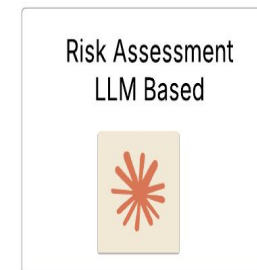
4. Data Summary Table

| Time (UTC) | Count (Int) | Entropy | Anisotropy | P1 Absolute | Delta (Count) |

[illegible]

| [Timestamp] | [Count] | [Entropy] | [Aniso] | [P1_Abs] | [Change] |

- Risk Assessment: <Status Verdict | Primary Trigger>
- Actionable Insight
- Trend Analysis
- Log Summary

[illegible]

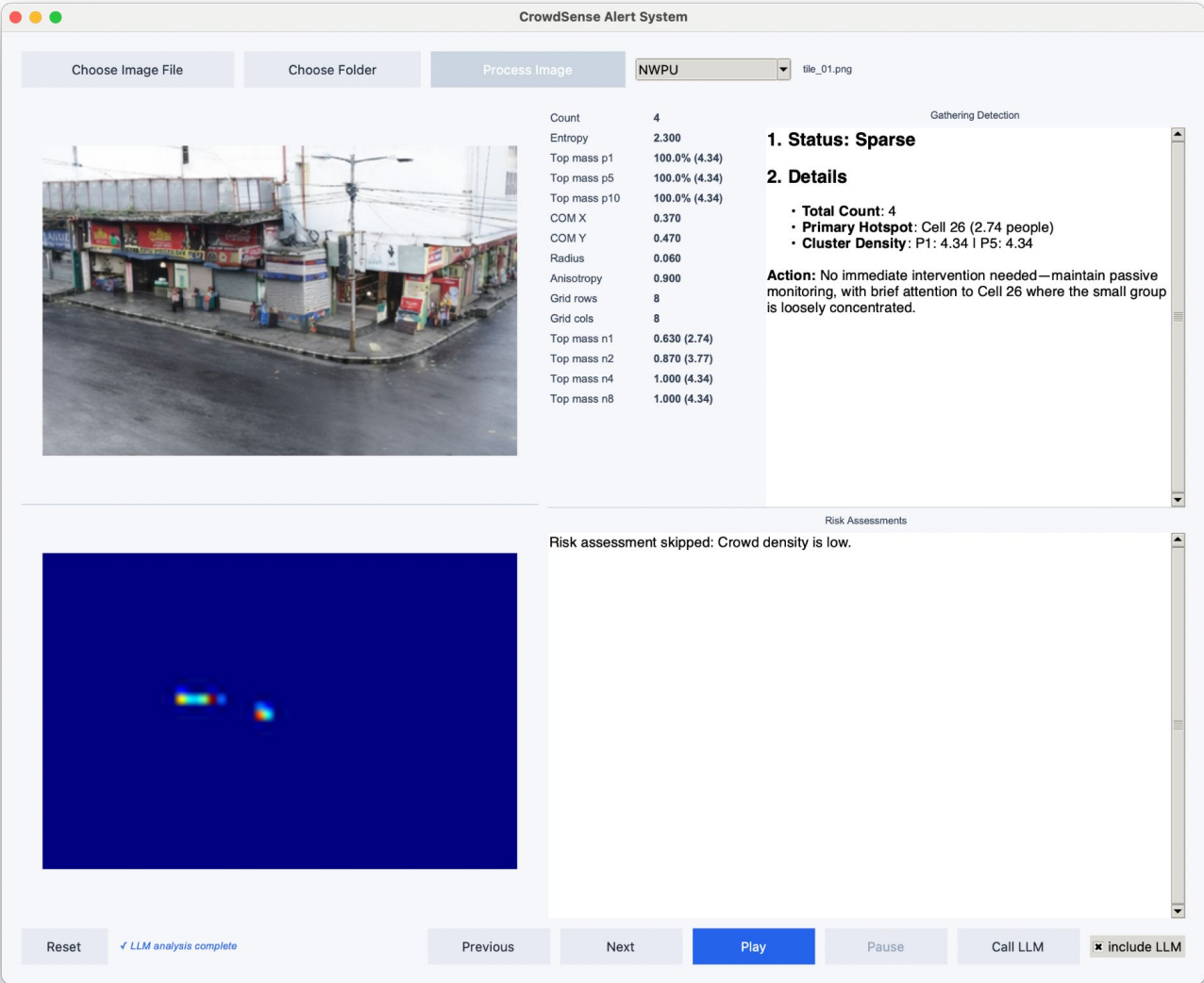
```
{
  'meta': {
    'image_name': "2.png",
    'model_name': "model_nwpu",
    'timestamp_utc': "2025-12-22T14:10:10.298056+00:00",
    'density_map_shape_hw': { "h": 112, "w": 150}
  },
  'feature_version': "dmcount_density_features_v1",
  'count': { 'count_float': 401.51, 'count_int': 402},
  'shape_norm': {
    'entropy': 7.262,
    'top_percent_mass': { "p1": 0.26, "p5": 0.74, "p10": 0.97},
    'spatial_norm': {
      'com_xy': [0.56, 0.48],
      'radius': 0.26
    },
    'anisotropy': 0.62
  },
  'grid': {
    'rows': 8,
    'cols': 8,
    'mass': [
      0.0960
      0.0144
      ...
      0.0026
      0.0028
    ],
    'top_mass': { "n1": 0.094, "n2": 0.159, "n4": 0.274, "n8": 0.456}
  }
}
```

```

    datasets
    stats
    model_nwpw_tile_01.json
    model_nwpw_tile_02.json
    model_nwpw_tile_03.json
    model_nwpw_tile_04.json
    model_nwpw_tile_05.json
    model_nwpw_tile_06.json
    model_nwpw_tile_07.json
    model_nwpw_tile_08.json
    model_nwpw_tile_09.json
    model_nwpw_tile_11.json
    model_nwpw_tile_12.json

```

CrowdSense Alert System - DEMO

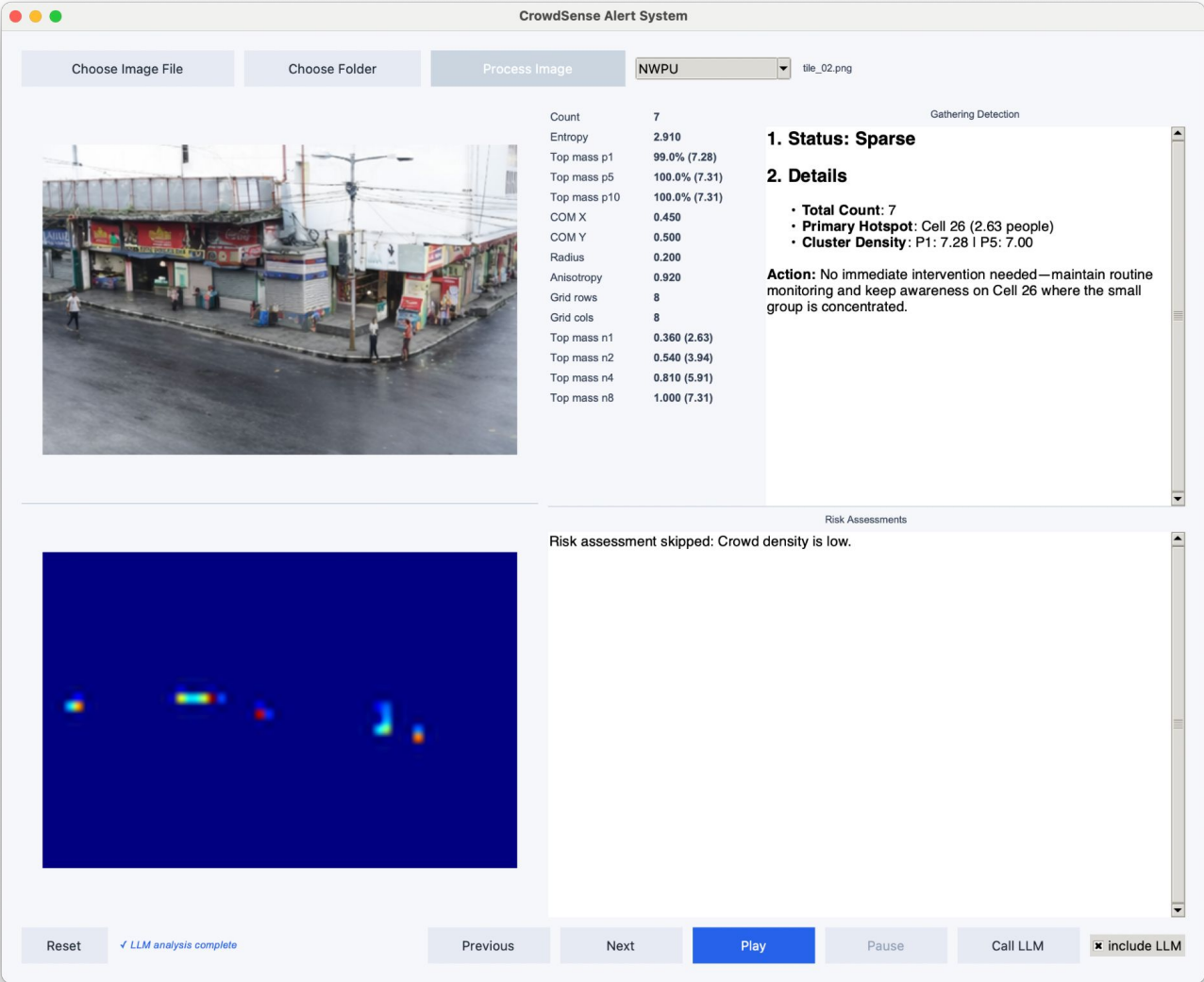


Status: Sparse

Action: No immediate intervention

Skipped - Low density

CrowdSense Alert System - DEMO

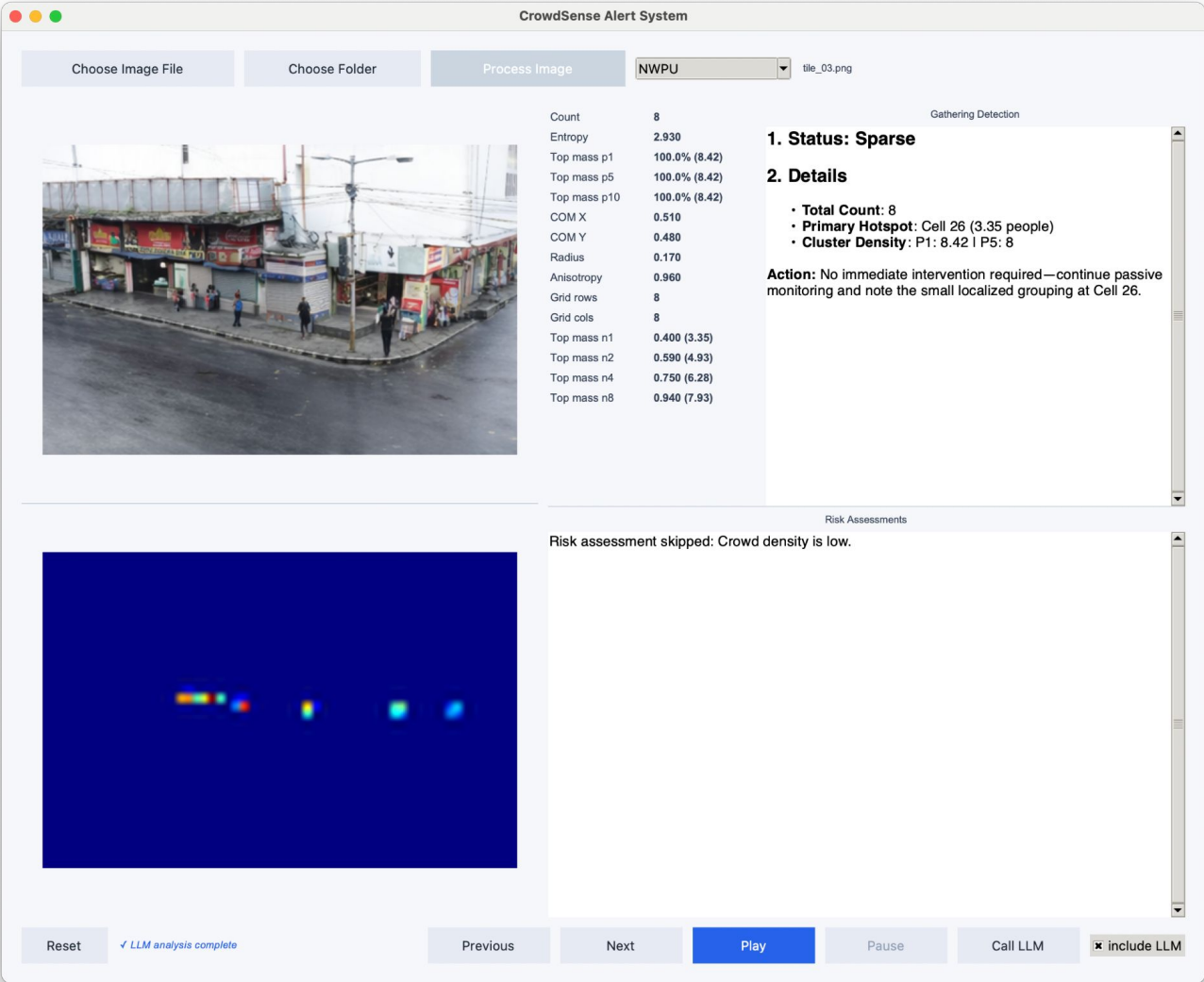


Status: Sparse

Action: No immediate intervention

Skipped - Low density

CrowdSense Alert System - DEMO

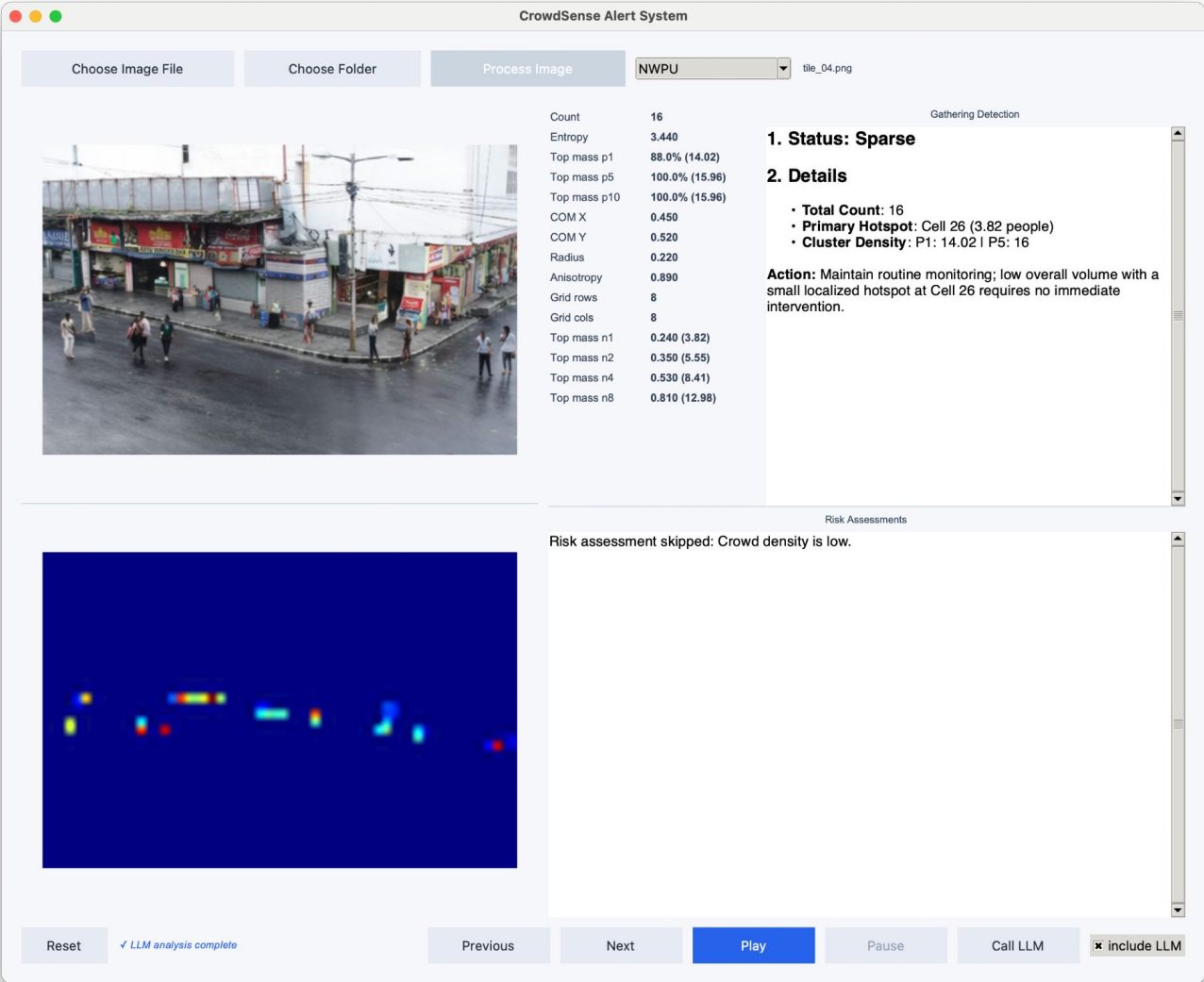


Status: Sparse

Action: No immediate intervention

Skipped - Low density

CrowdSense Alert System - DEMO

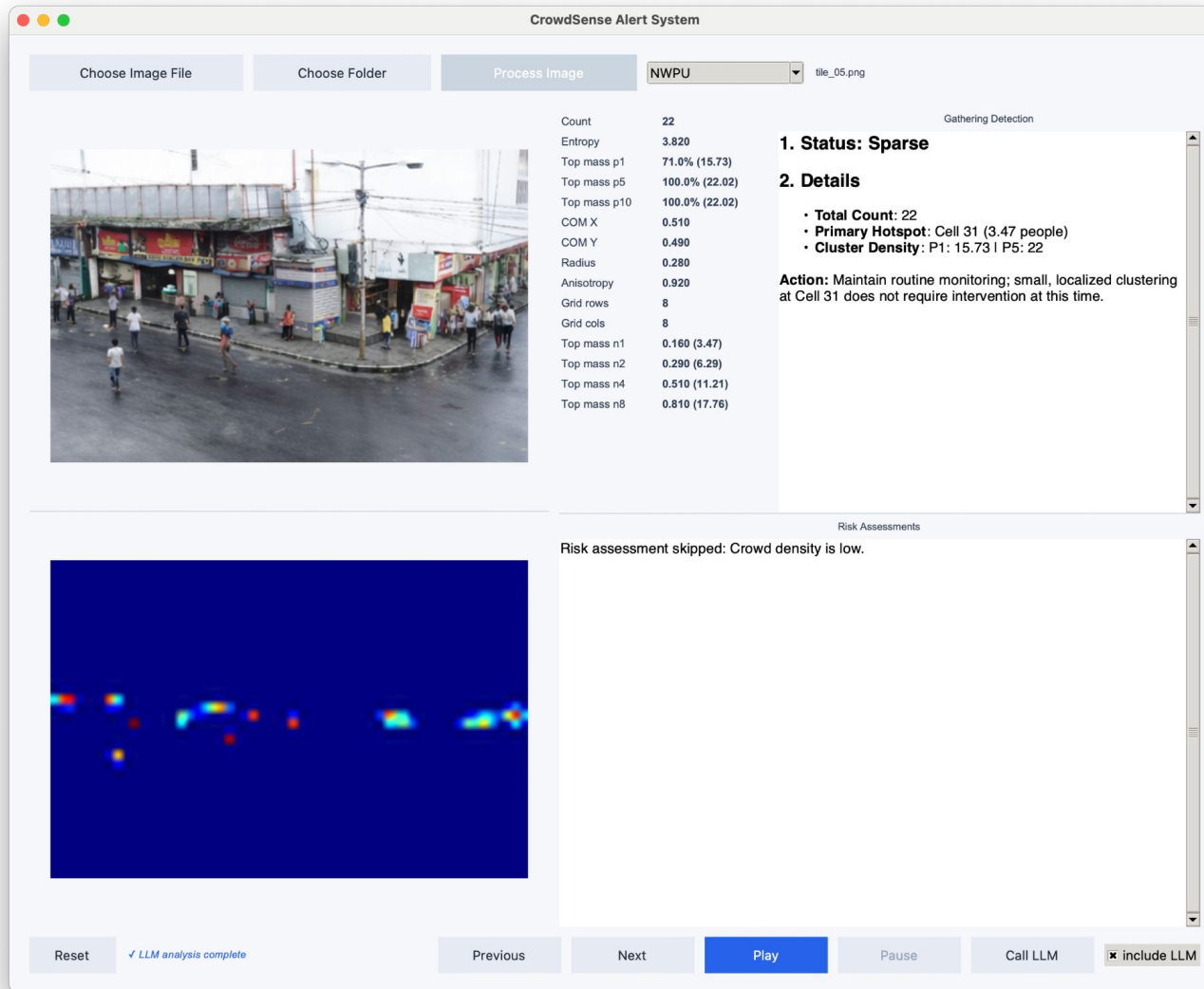


Status: Sparse

Action: Maintain routine monitoring
low overall volume

Skipped - Low density

CrowdSense Alert System - DEMO

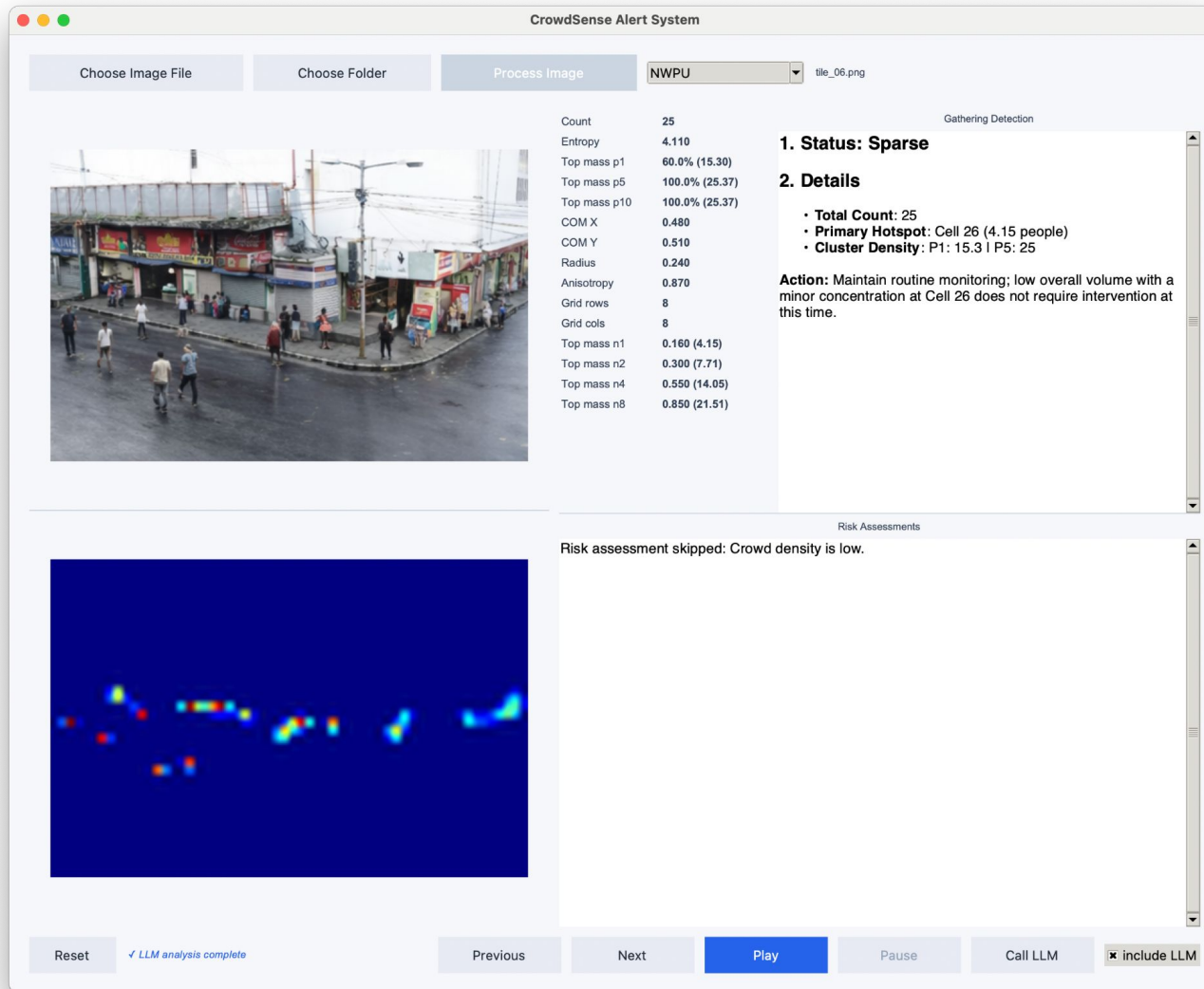


Status: Sparse

Action: Maintain routine monitoring
small localized clustering

Skipped - Low density

CrowdSense Alert System - DEMO

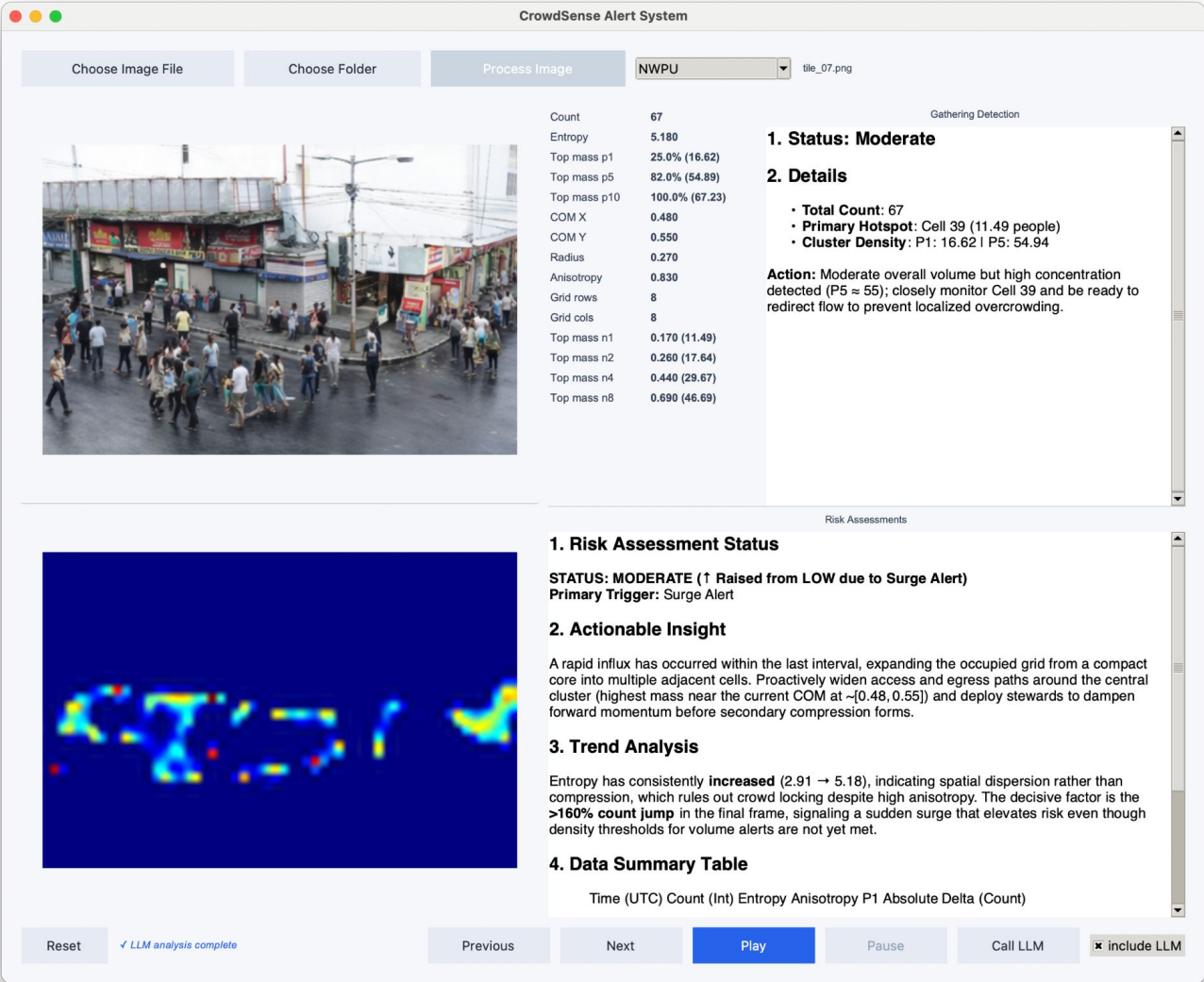


Status: Sparse

Action: Maintain routine monitoring
low overall with minor concentration

Skipped - Low density

CrowdSense Alert System - DEMO



Status: Moderate

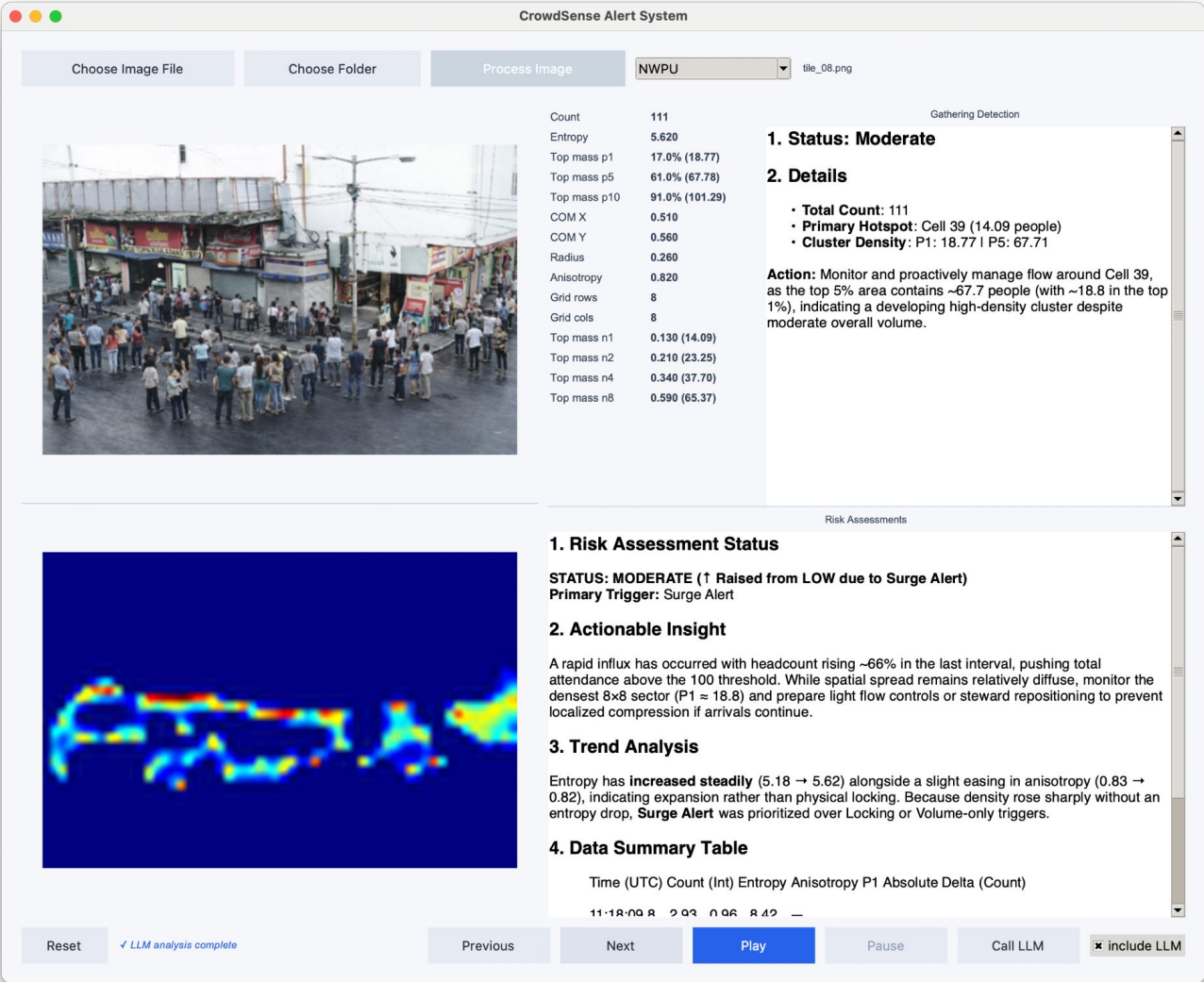
Action: Moderate overall volume, but high concentration detected (P5) be ready to redirect flow

Status: Moderate
raised from Low due to surge alert

Insight: A rapid influx has occurred within the last interval

Trend: Entropy has consistently increased indicating spatial dispersion rather than compression - which rules out crowd locking despite high anisotropy

CrowdSense Alert System - DEMO



Status: Moderate

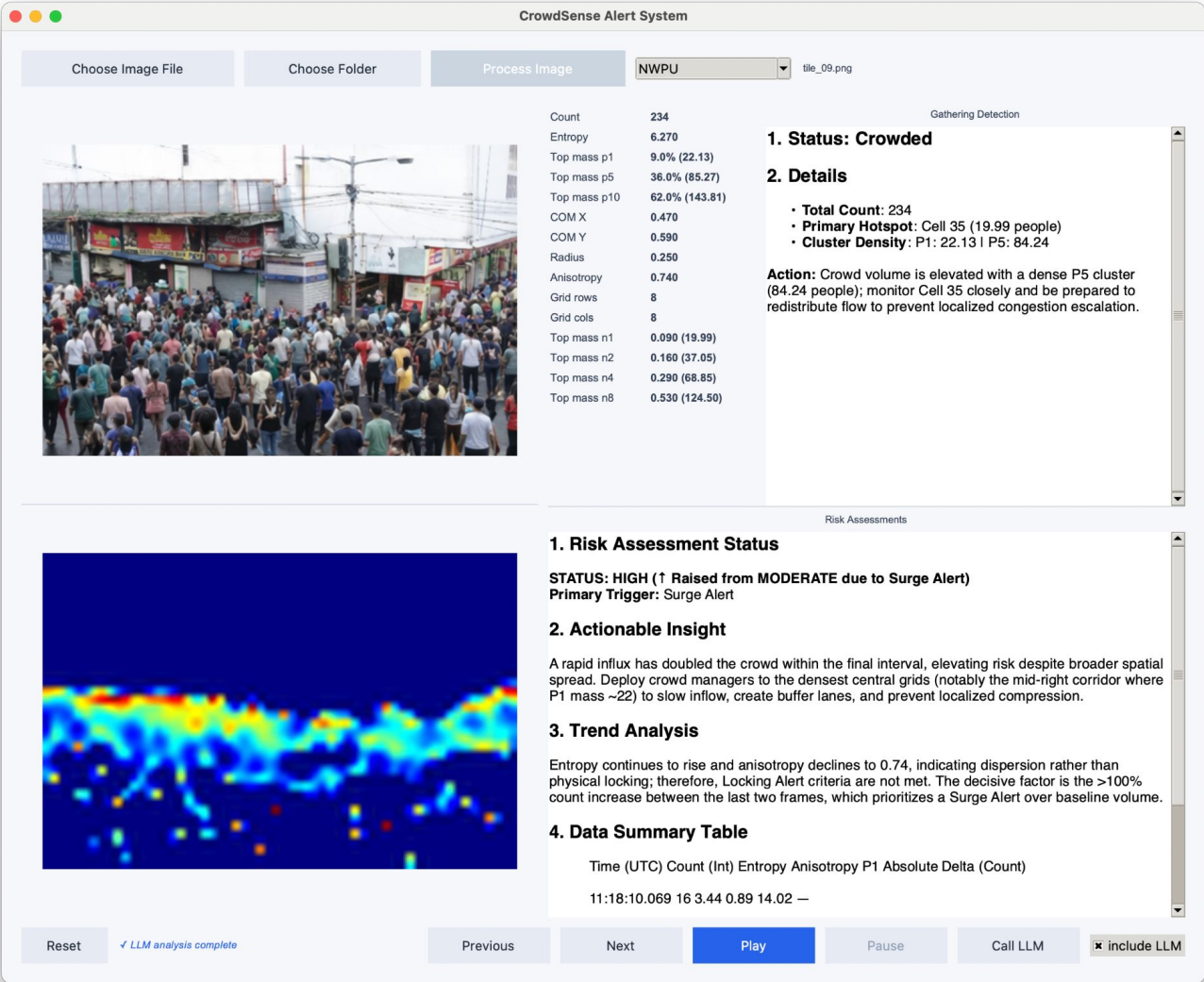
Action: Monitor and proactively manage flow..developing high density cluster despite moderate overall volume

Status: Moderate

Insight: rapid influx with headcount rising ~66% in the last interval

Trend: Entropy increase steadily, easing anisotropy indicating expansion rather than physical locking

CrowdSense Alert System - DEMO

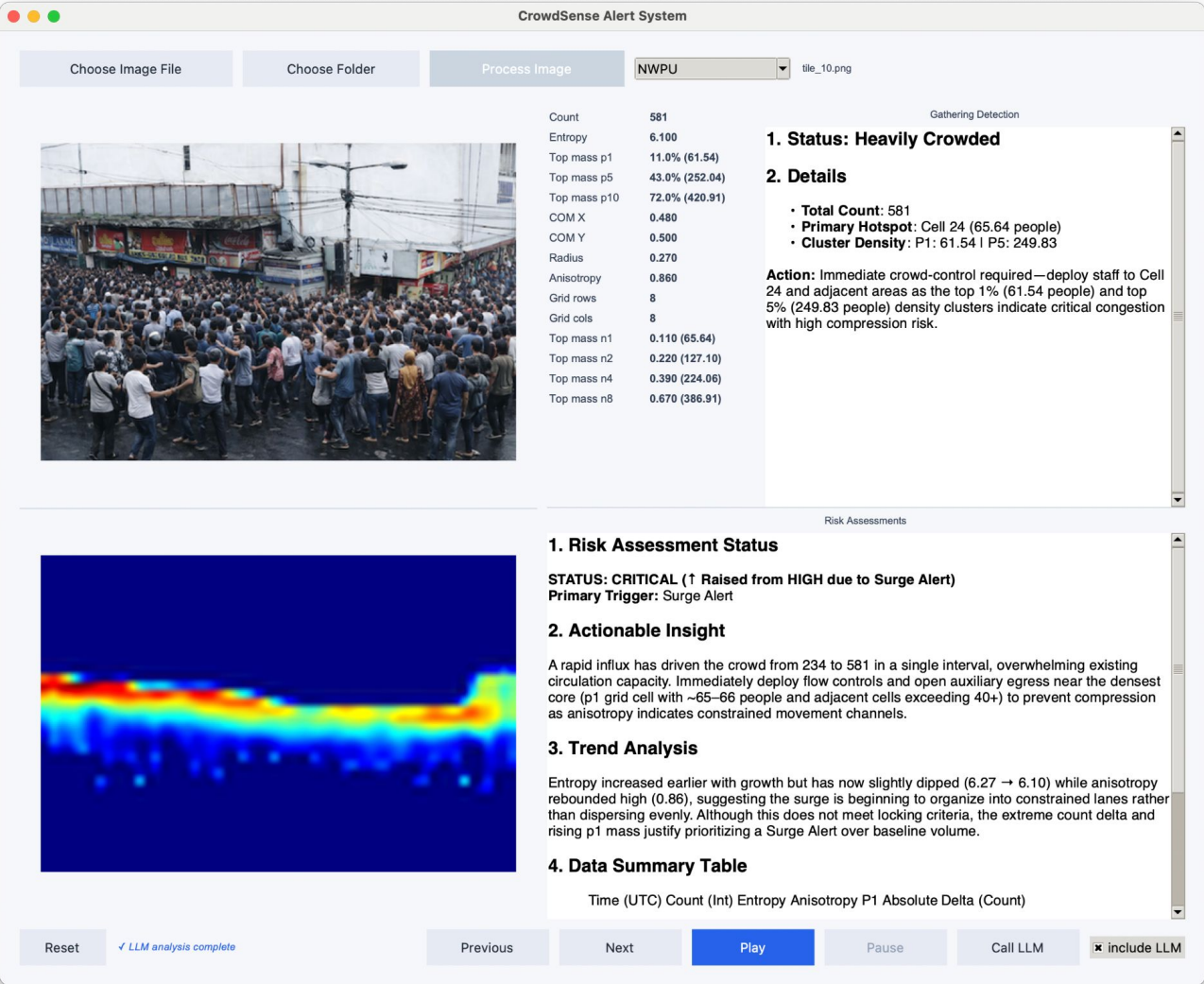


Status: Crowded

Action: Crowd volume is elevated with a dense P5 cluster.. be prepared to redistribute flow to prevent localized congestion escalation

Status: High (Raised From Moderate)
Insight: A rapid influx has doubled the crowd within the final interval
Trend: Entropy continues to rise, anisotropy declines, Locking Alert criteria are not met... prioritize a Surge Alert

CrowdSense Alert System - DEMO



Status: Heavily Crowded

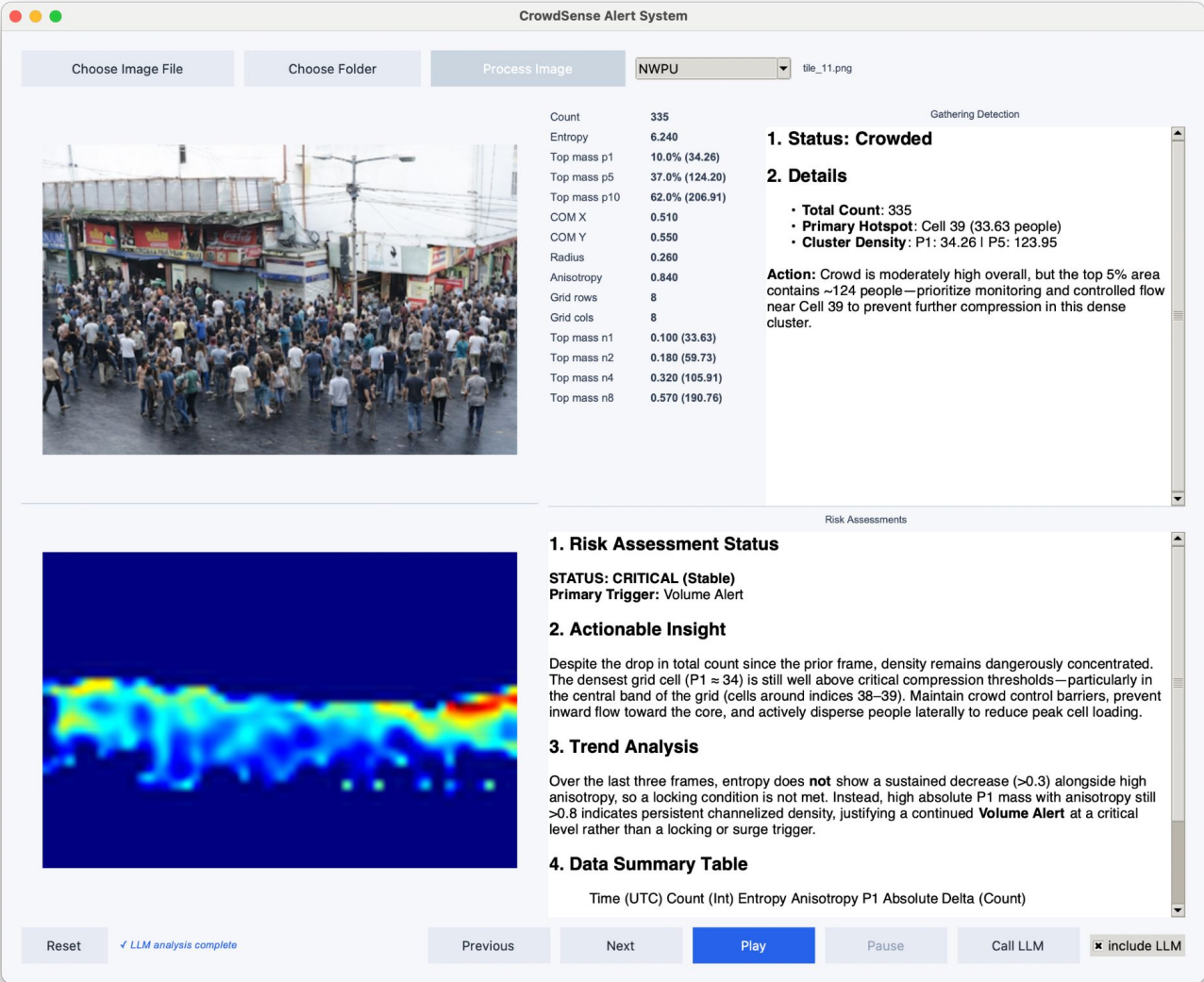
Action: Immediate crowd-control required, top 1% and 5% (249) indicate critical congestion with high compression risk

Status: Critical (Raised From High)

Insight - A rapid influx has driven the crowd in a single interval.. immediately deploy flow control

Trend: Entropy increased but has now slightly dipped while anisotropy rebounded high, suggesting the surge is beginning to organize into constrained lanes rather than dispersing evenly

CrowdSense Alert System - DEMO



Status: Crowded

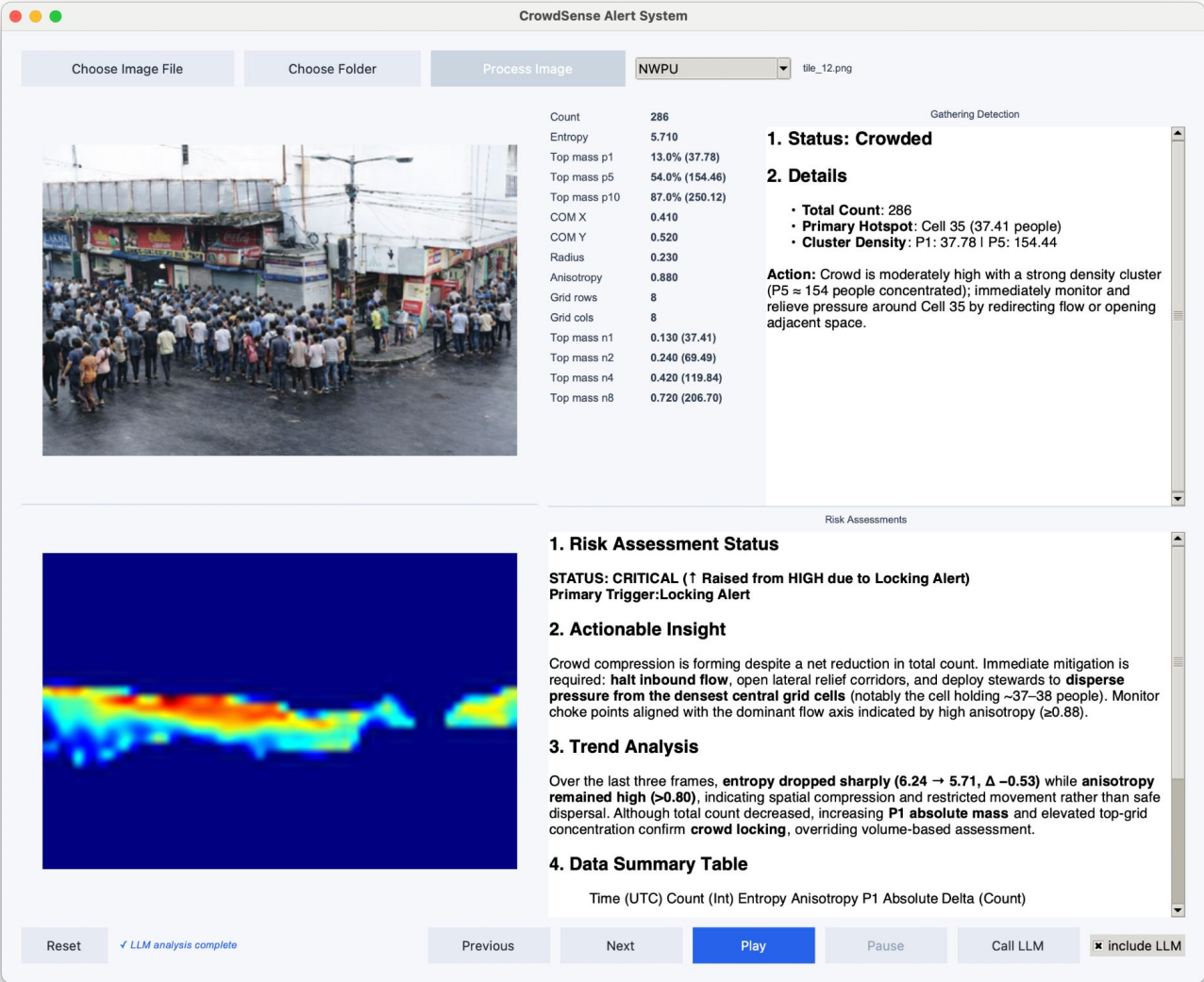
Action: Crowd is moderately high the top 5% contains ~124 people .. prioritize monitoring and controlled flow near cell 39 to prevent further compression in the dense cluster

Status: Critical (Stable)

Insight: Despite the drop in total count..density remain dangerously concentrated

Trend: over the last three frames, entropy does not show a sustained decrease along side high anisotropy..Volume Alert at a critical level

CrowdSense Alert System - DEMO



Status: Crowded

Action: Crowd is moderately high with a strong density cluster (P5 =154)

Status: Critical | Primary Trigger: Locking Alert

Insight: Crowd compression is forming despite a net reduction in total. immediate mitigation is required - halt inbound flow, open lateral relief corridors deploy stewards to disperse pressure

Trend: over the last three frames, entropy dropped sharply while anisotropy remain high..crowd locking

Challenges / Issues / Mitigations

- Frame Size
 - Performance Impact (DM-Count)
- Intense Configuration (Model | Camera View | Monitoring Area)
 - DM-Count
 - Pre-trained model type
 - Thresholds
 - Gathering Definition
 - Risk Assessments
- Real Time Activities
 - LLM single request might take up to 20 sec
- Cost
 - 100 requests ~ 1\$
- Mitigations
 - Image Preprocessing (size reduction)
 - Smart Caching
 - Local LLM